⚠ IMPORTANT — Support Scope

3DPTronics supports the VirtuaTilt controller hardware only. This guide is a *compendium* to help you connect your VirtuaTilt to third-party pinball software (VPX, Future Pinball + BAM, DOF/DOFLinx, PinEvent, PinUP Popper, etc.). We provide them as a courtesy to save you the hunt across multiple sites. **The information is best-effort** and kept reasonably current, but is **not** official documentation of those projects. **We do not provide official support for any third-party software referenced here** — for software-specific issues, refer to each project's own docs and community channels.

✅ Officially supported

Component	Coverage
🎮 **VirtuaTilt controller**	Wired and Wireless editions, all hardware components
💾 **Pinscape Pico firmware**	Updates, OTA flashing, recovery via Updater Tool
📋 **Bundled profiles**	All BT and USB profiles shipped in the redist
⚙ **GUIConfigTool**	Configuration, calibration, profile loading, output testing
🔧 **Wireless Kit**	Wireless hardware kit for existing VirtuaTilt owners
📦 **Accessories**	Anything shipped directly by 3DPTronics

❌ Not supported

Category	Examples	Where to get help
🎲 **VP Game engines**	VPX, Visual Pinball, Future Pinball, BAM	VPForums · VP Universe
🕹 **VP titles**	Pinball FX / FX2 / FX3 / VR / M, etc.	Zen Studios support
💥 **Feedback frameworks**	DOF (Direct Output Framework), DOFLinx	DOF on GitHub · DOFLinx wiki
🎨 **DMD / extras**	DMDExt, PinEvent V2, PuP-Packs / PUP-Stream	VP Universe
🚀 **Frontends**	PinUP Popper, Baller Installer, etc.	Popper forum
🎬 **Emulators / ROMs**	MAME, pinball ROMs	MAMEdev
💻 **System-level**	Windows, GPU/audio drivers, USB controllers	Microsoft / vendor support

🌐 Community help for software

Project	Community
Visual Pinball X	vpforums.org · vpuniverse.com
Future Pinball + BAM	VPU Future Pinball forum
Future Pinball + BAM AIO	VPU file 14807 — FP+BAM Essentials AIO
DOF	github.com/DirectOutput/DirectOutput
DOFLinx	doflinx.github.io/docs
Pinball FX family	Zen Studios Support
PinUP Popper	VPU PinUP Popper forum
MAME	mamedev.org

💡 **Why this matters:** Our team is small and our expertise is the VirtuaTilt hardware — we can't debug VPX scripts, flaky BAM installs, or broken DOF chains. Those communities have their own experts. The **AI chatbot on the wiki** can help connect VirtuaTilt with most of the software above; for deeper or vendor-specific issues, reach out to the appropriate community first.

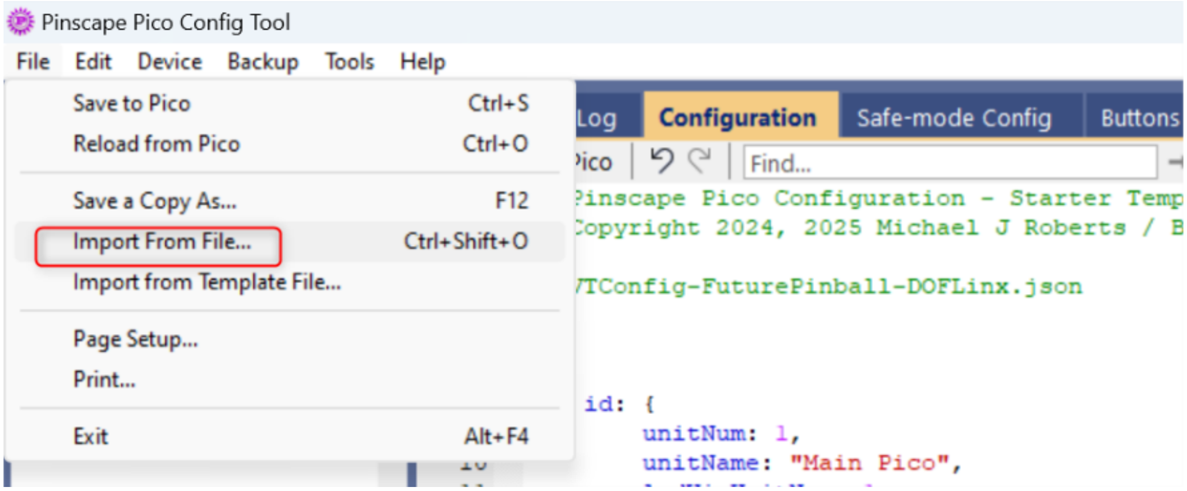
VIRTUATILT – PinballFX VR Setup

Following some basic instructions to have your new VirtuaTilt up and running with PinballFX VR.

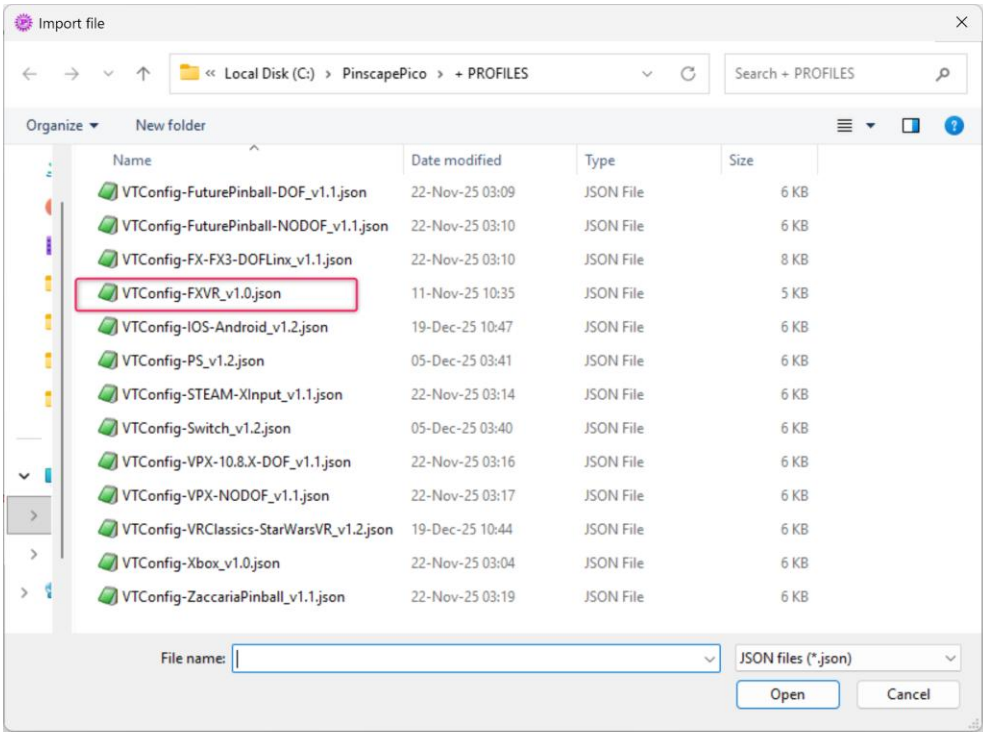
NOTE: You need an OTG adapter to directly connect your VirtuaTilt to Meta Quest (USB-A to USB-C)

PREREQUISITES:

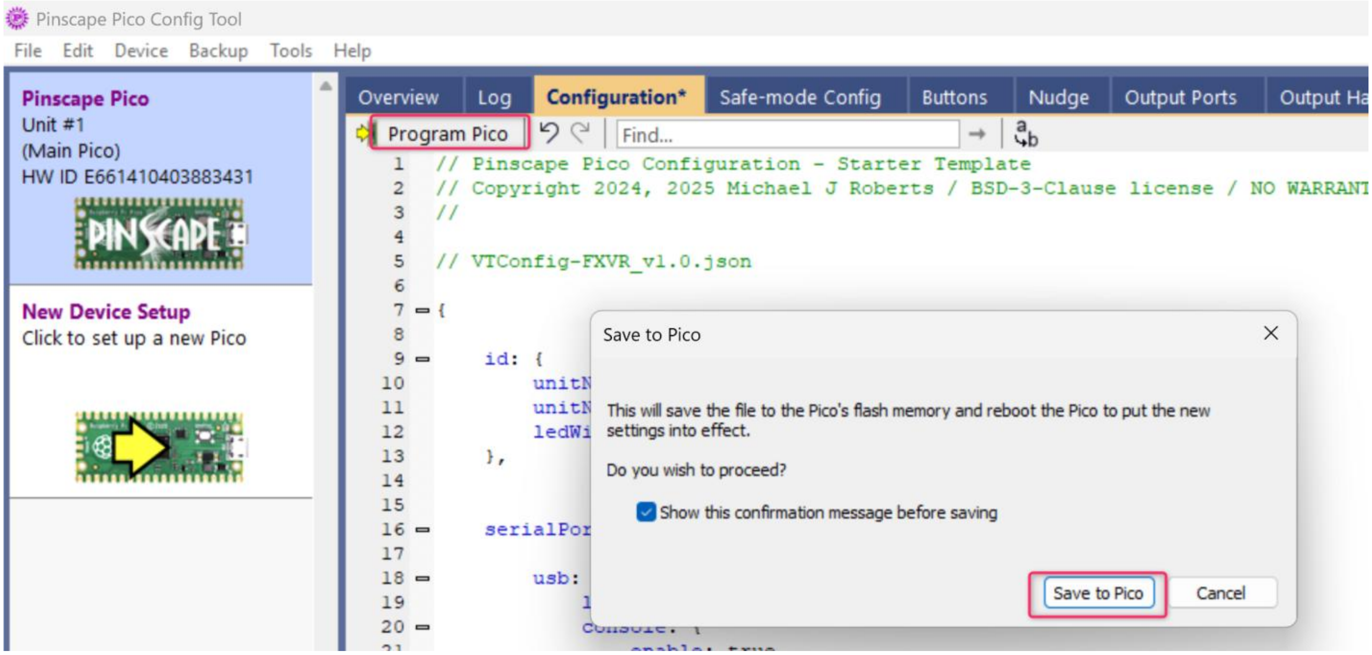
- **VTConfig-FXVR_v1.X.json** profile loaded in **Pinscape Pico Config Tool**


Pinscape Pico Config Tool
File
Edit
Device
Backup
Tools
Help
Save to Pico Ctrl+S
Reload from Pico Ctrl+O
Save a Copy As... F12
Import From File... Ctrl+Shift+O
Import from Template File...
Page Setup...
Print...
Exit Alt+F4
Log
Configuration
Safe-mode Config
Buttons
ico
Find...

```
inscape Pico Configuration - Starter Temp
opyright 2024, 2025 Michael J Roberts / B

TConfig-FuturePinball-DOFLinx.json


 id: {
     unitNum: 1,
     unitName: "Main Pico",
```


Import file
Local Disk (C:) › PinscapePico › + PROFILES
Search + PROFILES
Organize
New folder
File name:
JSON files (*.json)
Open
Cancel

Name	Date modified	Type	Size
VTConfig-FuturePinball-DOF_v1.1.json	22-Nov-25 03:09	JSON File	6 KB
VTConfig-FuturePinball-NODOF_v1.1.json	22-Nov-25 03:10	JSON File	6 KB
VTConfig-FX-FX3-DOFLinx_v1.1.json	22-Nov-25 03:10	JSON File	8 KB
VTConfig-FXVR_v1.0.json	11-Nov-25 10:35	JSON File	5 KB
VTConfig-IOS-Android_v1.2.json	19-Dec-25 10:47	JSON File	6 KB
VTConfig-PS_v1.2.json	05-Dec-25 03:41	JSON File	6 KB
VTConfig-STEAM-XInput_v1.1.json	22-Nov-25 03:14	JSON File	6 KB
VTConfig-Switch_v1.2.json	05-Dec-25 03:40	JSON File	6 KB
VTConfig-VPX-10.8.X-DOF_v1.1.json	22-Nov-25 03:16	JSON File	6 KB
VTConfig-VPX-NODOF_v1.1.json	22-Nov-25 03:17	JSON File	6 KB
VTConfig-VRClassics-StarWarsVR_v1.2.json	19-Dec-25 10:44	JSON File	6 KB
VTConfig-Xbox_v1.0.json	22-Nov-25 03:04	JSON File	6 KB
VTConfig-ZaccariaPinball_v1.1.json	22-Nov-25 03:19	JSON File	6 KB


Pinscape Pico Config Tool
File
Edit
Device
Backup
Tools
Help
Overview
Log
Configuration*
Safe-mode Config
Buttons
Nudge
Output Ports
Output Ha
Program Pico
Find...

Pinscape Pico
Unit #1
(Main Pico)
HW ID E661410403883431

New Device Setup
Click to set up a new Pico

```
// Pinscape Pico Configuration - Starter Template
// Copyright 2024, 2025 Michael J Roberts / BSD-3-Clause license / NO WARRANT
//

// VTConfig-FXVR_v1.0.json

{

    id: {
        unitN
        unitN
        ledWi
    },

    serialPor

        usb:
            l
        console: {
```

Save to Pico

This will save the file to the Pico's flash memory and reboot the Pico to put the new settings into effect.

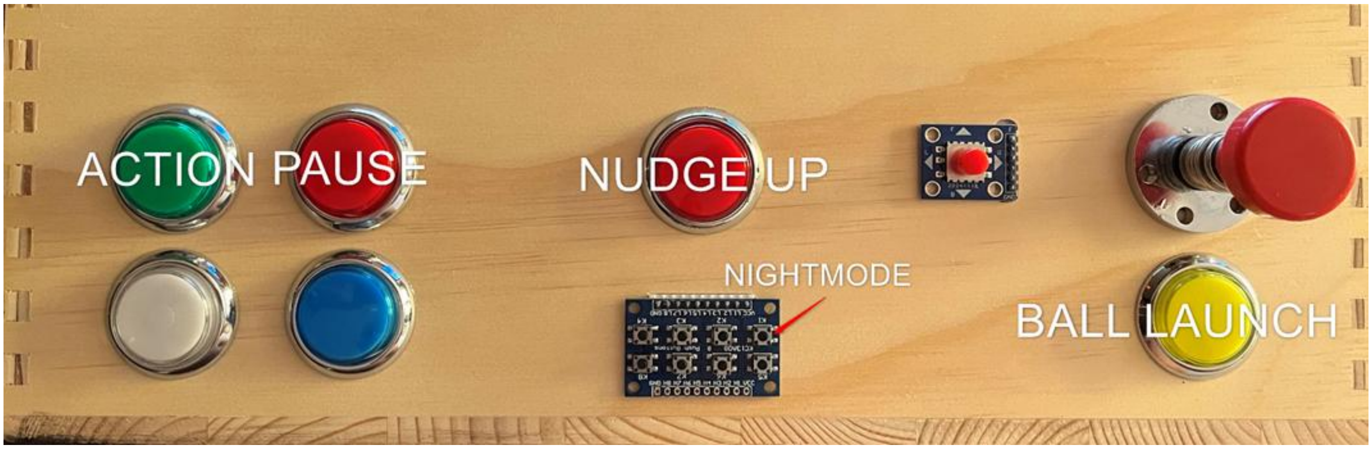
Do you wish to proceed?

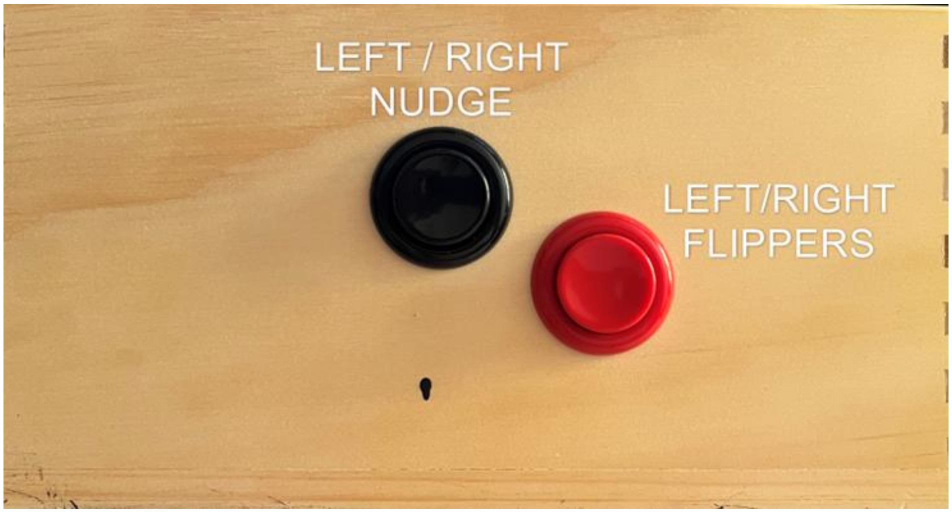
☑ Show this confirmation message before saving

Save to Pico
Cancel

Buttons Mapping

This is default button mapping for PinballFX VR:


ACTION
PAUSE
NUDGE UP
NIGHTMODE
BALL LAUNCH

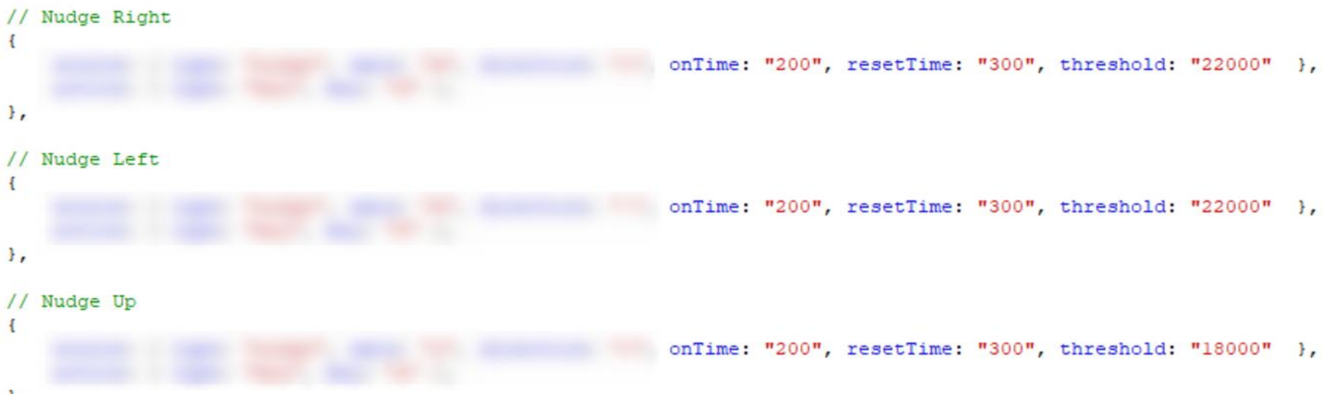

LEFT / RIGHT NUDGE
LEFT/RIGHT FLIPPERS

Nudge Adjustments

Nudge is already configured and ready to use. You can nudge with left/right buttons **OR / AND** internal accelerometer.

IMPORTANT NOTE: **PinballFX VR does not support analog nudge natively.** However, VirtuaTilt is able to translate digital inputs into accelerometer movements!

You can adjust parameters for accelerometer-based nudging in **Pinscape Pico Config Tool** under **Configuration** tab:



```
// Nudge Right
{
    [redacted] onTime: "200", resetTime: "300", threshold: "22000"  },
},

// Nudge Left
{
    [redacted] onTime: "200", resetTime: "300", threshold: "22000"  },
},

// Nudge Up
{
    [redacted] onTime: "200", resetTime: "300", threshold: "18000"  },
```

onTime

The amount of time to pulse the imaginary button, in milliseconds (1000 milliseconds equals one second). When the nudge axis reading goes over the threshold, the button reads as pressed for this interval.

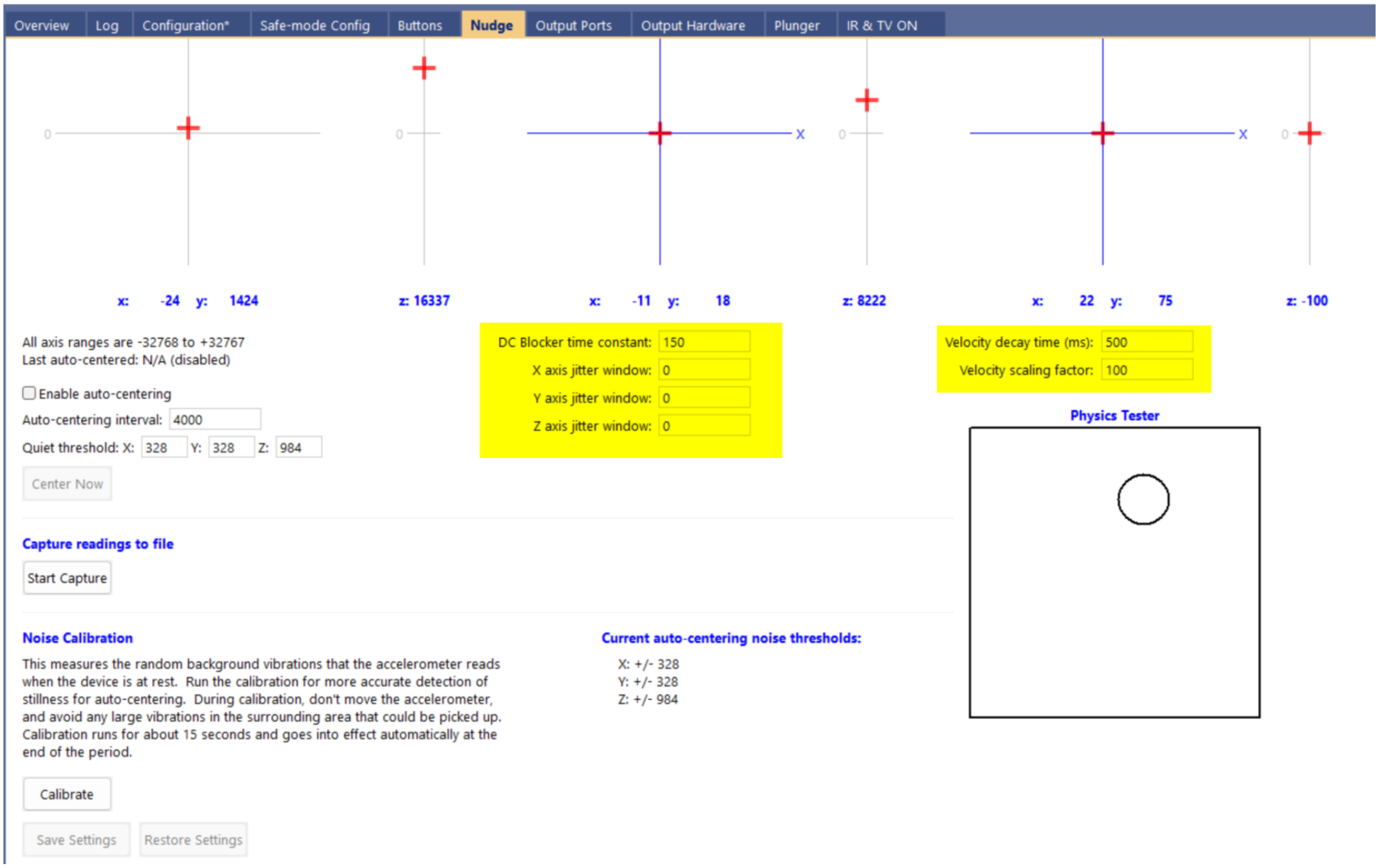
ResetTime

The reset time for the nudge event, in milliseconds (1000 milliseconds equals one second). Each time the button is triggered, it performs its action and then ignores the nudge input for this amount of time. This prevents the button from getting triggered repeatedly during a strong jolt to the cabinet.

Threshold

The minimum reading on the nudge axis to trigger the event, 0 to 32767.

VERY IMPORTANT NOTE: You need to find your right settings (according to your personal preferences) for nudging. You can adjust both Pinscape Pico accelerometer settings and your VP game settings until you find your perfect spot:


Overview
Log
Configuration*
Safe-mode Config
Buttons
Nudge
Output Ports
Output Hardware
Plunger
IR & TV ON
0
X
Physics Tester
Center Now
Start Capture
Calibrate
Save Settings
Restore Settings

x: -24 y: 1424

z: 16337

x: -11 y: 18

z: 8222

x: 22 y: 75

z: -100

All axis ranges are -32768 to +32767
Last auto-centered: N/A (disabled)

☐ Enable auto-centering

Auto-centering interval: 4000

Quiet threshold: X: 328 Y: 328 Z: 984

DC Blocker time constant: 150
X axis jitter window: 0
Y axis jitter window: 0
Z axis jitter window: 0

Velocity decay time (ms): 500
Velocity scaling factor: 100

Capture readings to file

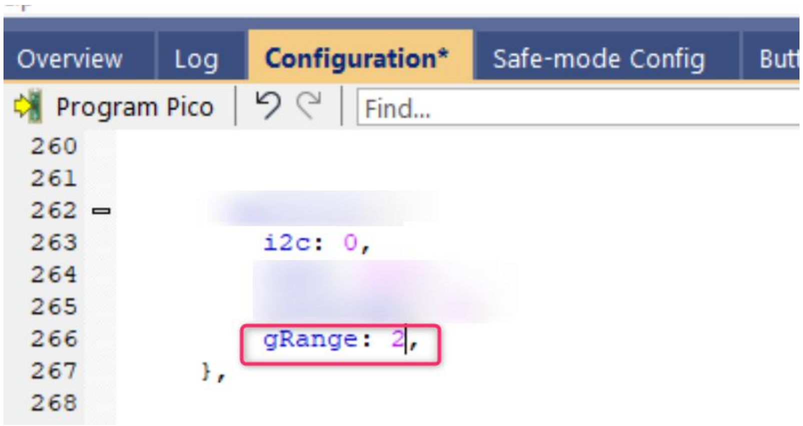
Noise Calibration

This measures the random background vibrations that the accelerometer reads when the device is at rest. Run the calibration for more accurate detection of stillness for auto-centering. During calibration, don't move the accelerometer, and avoid any large vibrations in the surrounding area that could be picked up. Calibration runs for about 15 seconds and goes into effect automatically at the end of the period.

Current auto-centering noise thresholds:

X: +/- 328
Y: +/- 328
Z: +/- 984

You can also increase accelerometer sensitivity by changing gRange value to 2 in any profile:


Overview
Log
Configuration*
Safe-mode Config
Butt
Program Pico
Find...

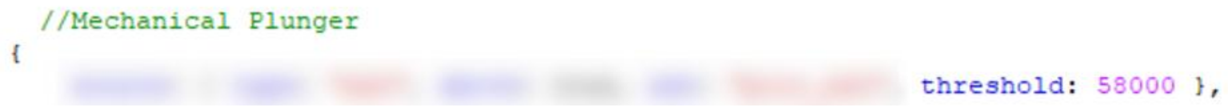
```
        i2c: 0,


        gRange: 2,
    },
```

Please refer to the official Pinscape Pico reference for additional information about accelerometer parameters: http://mjrnet.org/pinscape/PinscapePico/Help/NudgeWin.htm

Plunger Adjustments

IMPORTANT NOTE: **PinballFX VR does not support analog plunger natively; only ball launch button.** VirtuaTilt is able to translate a digital key press (ball launch button) into plunger movements, however it will never behave like a real analog plunger as in Pinball VR Classics or StarWars Pinball VR for example.

Plunger is already configured and ready to use. You can launch ball with button **OR / AND** mechanical plunger. You can adjust parameters for mechanical plunger in **Pinscape Pico Config Tool** under **Configuration** tab:



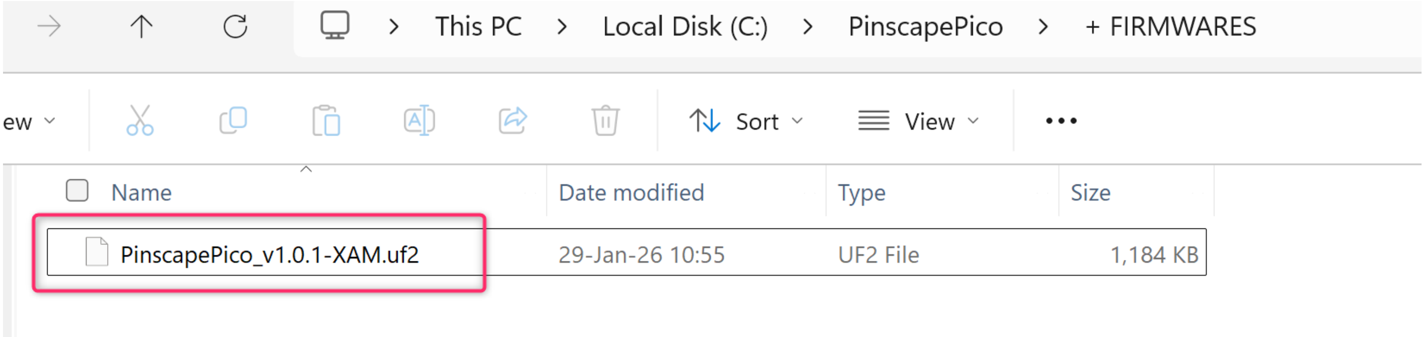
```
  //Mechanical Plunger
{
    [redacted] threshold: 58000 },
```

`Threshold`

Sets the analog reading at which the button will trigger, using the "normalized" range of 0 to 65535. The Pinscape software internally maps every device's native range to the same 0..65535 scale, where 0 corresponds the minimum voltage that the device can register, and 65535 corresponds to the maximum voltage.

X-Arcade Mode

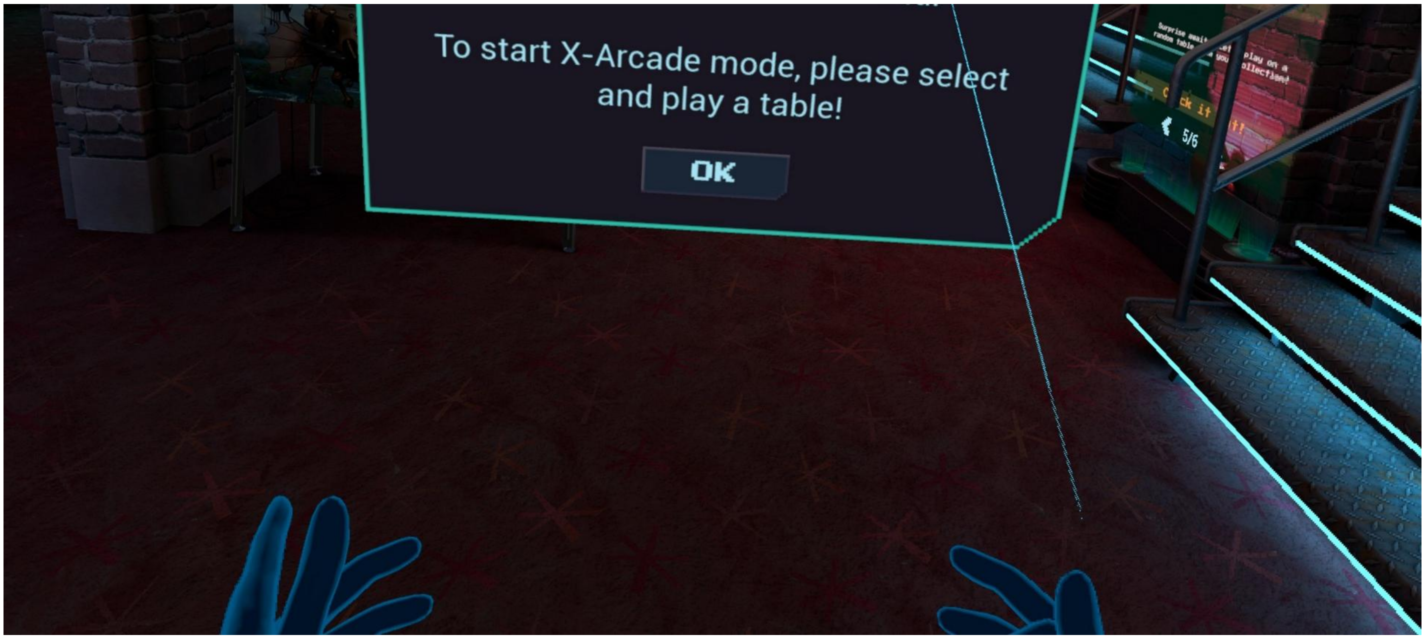
To activate X-Arcade Mode in PinballFX VR you need to download a custom firmware (**PinscapePico_v1.X.X-XAM.uf2**) available in shared folder and flash it to your Pico as described in **MAIN SETUP GUIDE – APPENDIX A**:



☐ Name	Date modified	Type	Size
PinscapePico_v1.0.1-XAM.uf2	29-Jan-26 10:55	UF2 File	1,184 KB

You also need to import a new profile in Pinscape Pico config tool: **VTConfig-FXVR_XAM_v1.X.json**

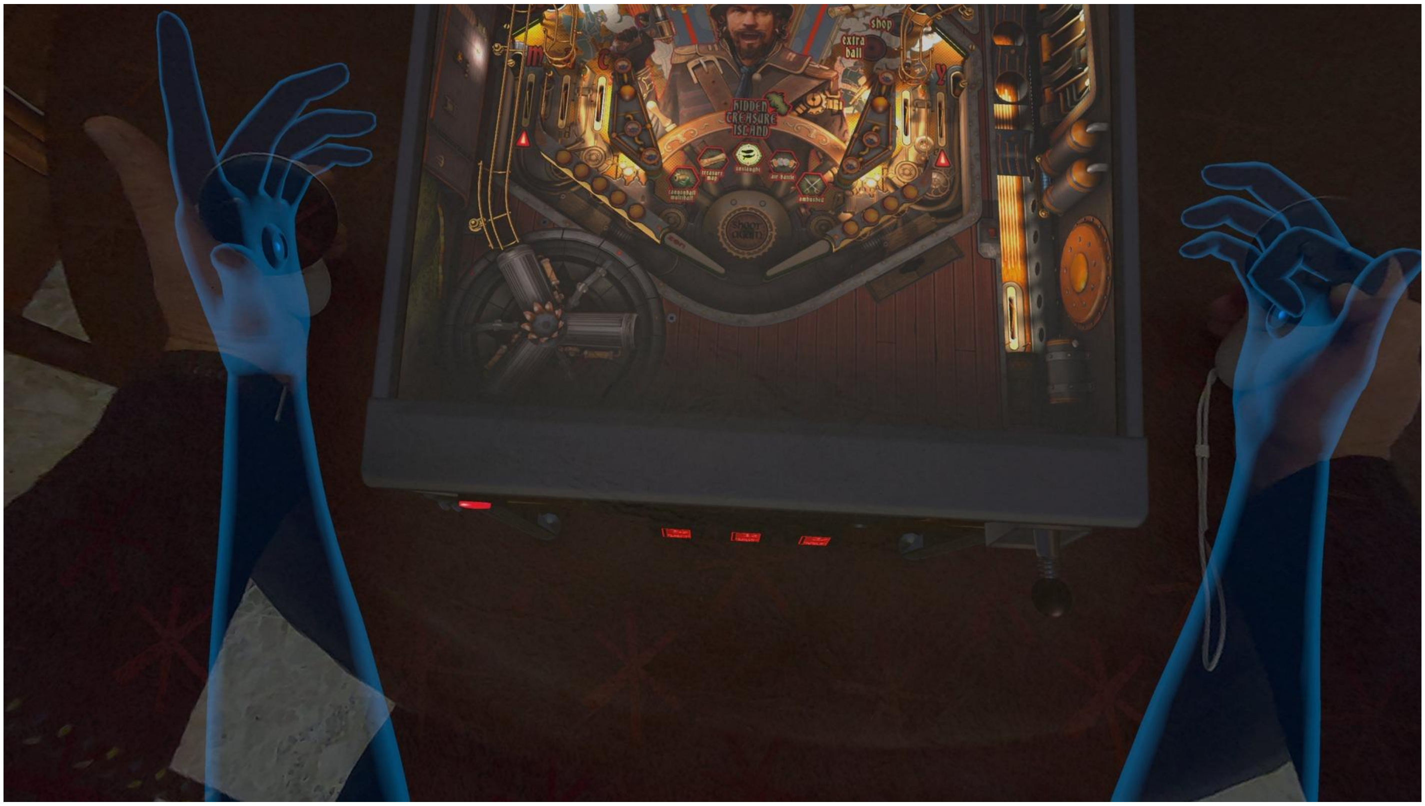
When you load the game, X-Arcade mode is automatically activated:


To start X-Arcade mode, please select and play a table!

Now you can benefit from new in-game features:

- Use left/right flipper buttons to navigate menus (**Play again /Exit table**)
- Recenter table view by using Meta button on right Quest controller
- **Use digital plunger steps to behave like an analog plunger!**

To activate this game mode, you need to grab your Meta Quest controllers, align them vertically to left and right flipper buttons and then press left and right flipper buttons simultaneously:



Experience the full potential of VirtuaTilt with **PinballFX VR**!